

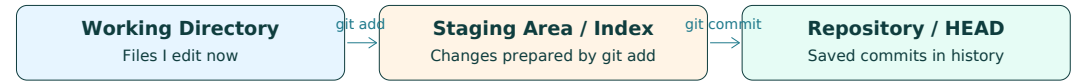
Git & Linux Cheat Sheet

Personal A4 one-page reference for terminal basics, Git workflow commands, and the Git three-tree model.

1) Basic Linux Commands

Command	Sample usage	What it does
pwd	pwd	Shows current folder.
ls	ls -la	Lists files; -l details, -a hidden.
cd	cd Documents	Moves to another folder.
mkdir	mkdir project	Creates a new folder.
touch	touch index.html	Creates an empty file.
cp	cp notes.txt backup.txt	Copies files; -r for folders.
mv	mv old.txt new.txt	Moves or renames.
rm	rm draft.txt	Deletes files; -r for folders.
cat	cat README.md	Prints file contents.
nano	nano index.html	Edits text in terminal.
clear	clear	Clears the screen.
man	man ls	Opens manual/help page.

3) Git Three-Tree Architecture



- Working Directory:** Your real files on disk. This is where edits happen.
- Staging Area / Index:** The waiting area for the exact changes you plan to commit.
- Repository / HEAD:** The saved Git history. HEAD points to your current commit.
- Fast checks:** git status = overall state; git diff = unstaged changes; git diff --staged = staged changes.

Memory hook: edit → stage → commit. Check before committing, then write clear commit messages.

2) Basic Git Commands

Command	Sample usage	Description in my words	Common options / flags
git init	git init git init my-site	Starts a new Git repository.	-b main first branch; --bare no working files; -q quiet
git restore	git restore index.html git restore --staged app.js	Undo uncommitted file edits or unstage files.	--staged unstage; --worktree restore file; -s choose source
git status	git status git status -sb	Shows changed, staged, and untracked files.	-s short; -b branch info; -u untracked files
git branch	git branch git branch feature-login	Lists, creates, renames, or deletes branches.	-a all; -d delete; -m rename; -v last commit
git add	git add index.html git add .	Stages selected changes for the next commit.	-A all changes; -p choose parts; -n dry run
git checkout / git switch	git switch main git switch -c feature git checkout -- old.txt	Moves between branches; checkout can restore paths.	switch -c create; checkout -b create; -- path restore
git commit	git commit -m "Add homepage"	Saves staged changes as a new history snapshot.	-m message; -a tracked files; --amend edit last commit
git merge	git merge feature-login	Combines another branch into the current branch.	--abort cancel; --no-ff merge commit; --squash one change
git log	git log --oneline --graph	Shows commit history and change trail.	--oneline compact; --graph visual; -p patches; --stat file stats
git tag	git tag v1.0 git tag -a v1.0 -m "First release"	Labels an exact commit, often for versions/releases.	-a annotated; -m message; -d delete; -l list
git diff	git diff git diff --staged	Shows line-by-line differences before committing.	--staged staged changes; --stat summary; -U<n> context
git clone	git clone https://github.com/user/repo.git	Downloads an existing repository to the computer.	--depth 1 shallow; -b branch; <dir> folder; -q quiet

Tip: Commit small, related changes with clear messages; use branches when trying features or experiments.

Content checked against built-in git command help output.